EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

d-BLOCK ELEMENTS

1. For a catalyst which condition is not essential **:**
(1) Variable oxidation states
(2) High ionisation energy
(3) Vacant orbitals
(4) Free valency on the surface

DB0001

2. To which of the following series the transition element from Z = 39 to Z = 48 belong :
(1) 3d series (2) 4d series
(3) 5d series (4) 6d series

DB0002

3. $KMnO_4$ on treatment with conc. H_2SO_4 forms a compound (X) which decomposes explosively on heating forming (Y). The (X) and (Y) are respectively
(1) Mn_2O_7, MnO_2 (2) Mn_2O, Mn_2O_3
(3) $MnSO_4$, Mn_2O_3 (4) Mn_2O_3, MnO_2

DB0003

4. Which pair of ions is colourless **:**
(1) Mn^{3+}, Co^{3+} (2) Fe^{3+}, Cr^{3+}
(3) Zn^{2+}, Sc^{3+} (4) Ti^{2+}, Cu^{2+}

DB0004

5. $(NH_4)_2Cr_2O_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is **:**
(1) Cr_2O_3 (2) CrO_2 (3) Cr_2O_4 (4) CrO_3

DB0005

6. $Cr_2O_7^{2-} \underset{Y}{\overset{X}{\rightleftharpoons}} 2CrO_4^{2-}$, X and Y are respectively
(1) X = OH^-, Y = H^+ (2) X = H^+, Y = OH^-
(3) X = OH^-, Y = H_2O_2 (4) X = H_2O_2, Y = OH^-

DB0006

7. The higher oxidation states of transition elements are found to be the combination with A and B, which are**:**
(1) F, O (2) O, N
(3) O, Cl (4) F, Cl

DB0008

8. Magnetic moment of x^{n+} is $\sqrt{24}$ B.M. Hence No. of unpaired electron and value of 'n' respectively. (Atomic number = 26)
(1) 4 , 3 (2) 3 , 5
(3) 4 , 2 (4) 4 , 1

DB0009

9. The product of oxidation of I^- with MnO_4^- in weak alkaline medium is :
(1) IO_3^- (2) I_2 (3) IO^- (4) IO_4^-

DB0010

10. Which of the following oxide of chromium is amphoteric in nature
(1) CrO (2) Cr_2O_3
(3) CrO_3 (4) CrO_5

DB0012

11. Compared to Cu^{2+} having $3d^9$ configuration, Cu^+ having $3d^{10}$ configuration (aq. solution) :-
(1) Is more stable
(2) Is equally stable
(3) Is less stable
(4) Stability depends upon nature of copper salt

DB0013

12. CrO_3 is red orange in colour. The nature of oxide is:-
(1) Acidic (2) Basic
(3) Amphoteric (4) Neutral

DB0015

13. Maximum spin magnetic moment is shown by :
(1) d^5 (2) d^6
(3) d^7 (4) d^8

DB0017

14. Disproportion can be shown by
(1) MnO_4^{2-} in acidic medium
(2) Cu^+ in aqueous medium
(3) Cl_2 in alkaline medium
(4) All of these

DB0018

15 Which of the following configurations of 3d series exhibits the largest number of oxidation states –
(1) [Ar] $3d^8\ 4s^2$ (2) [Ar] $3d^{10}\ 4s^1$
(3) [Ar] $3d^5\ 4s^2$ (4) [Ar] $3d^7\ 4s^2$

DB0024

16 Which of the following pair of ions has same value of "spin-only" magnetic moment
(1) Cu^+, Cu^{2+} (2) Co^{3+}, Fe^{2+}
(3) Ti^{2+}, V^{2+} (4) Sc^{2+}, Zn^{+2}

DB0026

17. Which of the following transition metal do not form amalgam with Hg :-
(1) Fe (2) Co
(3) Ni (4) All of the above
DB0145

18. Highest fluoride and oxide of Mn are respectively:-
(1) MnF_7, Mn_2O_7 (2) MnF_5, Mn_2O_7
(3) MnF_4, Mn_2O_7 (4) MnF_6, Mn_2O_7
DB0146

19. Which is not a property of intestitial compound ?
(1) Non stiochiometric
(2) Harder than pure metal
(3) Chemically inert
(4) Having low melting point
DB0147

20. Which halide of copper does not exist ?
(1) CuF_2 (2) $CuCl_2$ (3) Cu_2I_2 (4) CuI_2
DB0148

21. Which order is correct for oxidising power ?
(1) $Cr_2O_7^{-2} < VO_2^+$ (2) $VO_2^+ > MnO_4^-$
(3) $VO_2^+ < Cr_2O_7^{-2}$ (4) $Cr_2O_7^{-2} > MnO_4^-$
DB0149

22. Permangnate titrations in presence of Hydrochloric acid are not satisfacoty because :-
(1) HCl is a weak acid
(2) HCl is a volatile acid
(3) $KMnO_4$ will oxidise HCl into Cl_2
(4) $KMnO_4$ will show disproportionation in presence of HCl
DB0150

23. Which compound is used in manufacturing of $K_2Cr_2O_7$?
(1) Na_2CrO_4 (2) CrO_5
(3) $FeCr_2O_4$ (4) CrO_3
DB0151

24. Which of the following does not give chromyl chloride test ?
(1) NaCl (2) KCl (3) NH_4Cl (4) $HgCl_2$
DB0152

25. Which is correct match
(1) Brass ; Cu + Zn
(2) Bronze ; Cu + Sn
(3) Stainless steel ; Cr + Ni
(4) All of the above
DB0153

26. Which of the following 3d series metal has positive value of $E^\circ(M^{+2}/M)$?
(1) Zn (2) Cu (3) Mn (4) Cr
DB0154

f-BLOCK ELEMENTS

27. Which of the following actinide element does not found in nature ?
(1) Th_{90} (2) Pa_{91}
(3) U_{92} (4) Lr_{103}
FB0155

28. Which element does not show +4 oxidation state:-
(1) Ce (2) Ti (3) Pt (4) Eu
FB0156

29. Which of the following ion is diamagnetic and colourless ?
(1) La^{+3} (2) Lu^{+3}
(3) Zn^{+2} (4) All of the above
FB0157

30. The general electronic configuration of lanthanoid elements is :
(1) [Xe] $4f^{0-14}$ $5d^{0-1}$ $6s^2$
(2) [Xe] $4f^{0-14}$ $5d^{1-2}$ $6s^1$
(3) [Xe] $4f^{0-14}$ $5d^{0-1}$ $6s^{1-2}$
(4) None of these
FB0030

31. Cerium can show the oxidation state of +4 because
(1) it resemble alkali metals
(2) it has very low value of I.E.
(3) of its tendency to attain noble gas configuration of xenon
(4) of its tendency to attain $4f^7$ configuration
FB0031

32. The highest oxidation state shown by actinides is:
(1) +6 (2) +7 (3) +5 (4) +4
FB0033

33. The electronic configuration of gadolinium (At. No. 64) is :
(1) [Xe] $4f^75d^16s^2$ (2) [Xe] $4f^85d^06s^2$
(3) [Xe] $4f^85d^16s^1$ (4) [Xe] $4f^75d^06s^2$
FB0034

34. Which oxidation state is not shown by lanthanoids?
(1) +2 (2) +3
(3) +4 (4) +6
FB0158

35. Which of the following species acts as a reducing agent ?

(1) Ce^{+4} (2) Lu^{+3} (3) Eu^{+2} (4) Ce^{+3}

FB0159

36. Which of the following pair have almost similar size?

(1) Ti_2 and Zr_{40} (2) Nb_{41} and Ta_{73}

(3) Y_{39} and La_{57} (4) Ca_{20} and Ir_{31}

FB0160

37. An increase in both atomic and ionic radii with atomic number occurs in any group of the periodic table. In accordance of this the ionic radii of Ti (IV) and Zr (IV) ions are 0.68 Å and 0.74 Å respectively but for Hf (IV) ion the ionic radius is 0.75 Å, which is almost the same as that for Zr (IV) ion. This is due to :-

(1) greater degree of covalency in compounds of Hf^{4+}

(2) Lanthanide contraction

(3) Difference in the co-ordination number of Zr^{+4} and Hf^{+4} in their compounds

(4) Actinide contraction

FB0039

38. When lanthanoid element is heated with sulphur then compound obtained is :-

(1) LnS_2 (2) Ln_2S_3 (3) Ln_2S (4) Ln_3S_4

FB0161

39. Formula of halide of lanthanoid may be

(1) LnX_3 (2) LnX_5

(3) LnX_2 (4) LnX_6

FB0162

40. What is the composition of misch metal alloy?

(1) Lanthanoid metal (95%) + Iron (5%)

(2) Cu (69%) + Zn (31%)

(3) Al (20%) + Cu (80%)

(4) Cr (50%) + Zn (30%) + Cu (20%)

FB0163

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**2**	**1**	**3**	**1**	**1**	**1**	**3**	**1**	**2**	**3**	**1**	**1**	**4**	**3**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**4**	**3**	**4**	**4**	**3**	**3**	**3**	**4**	**4**	**2**	**4**	**4**	**4**	**4**
Que.	31	32	33	34	35	36	37	38	39	40					
Ans.	**3**	**2**	**1**	**4**	**3**	**2**	**2**	**2**	**1**	**1**					

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EXERCISE-III (Analytical Questions) — Master Your Understanding

1. The correct order of ionic radii of Y^{3+}, La^{3+}, Eu^{3+} and Lu^{3+} is

(1) $Lu^{3+} < Eu^{3+} < La^{3+} < Y^{3+}$

(2) $La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}$

(3) $Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}$

(4) $Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$

FB0096

2. The aqueous solution containing which one of the following ions will be colourless :-

[Atomic n : Sc = 21, Fe = 26, Ti = 22, Mn = 25]

(1) Fe^{2+} (2) Mn^{2+}

(3) Ti^{3+} (4) Sc^{3+}

DB0098

3. The lanthanide contraction is responsible for the fact that

(1) Zr and Y have about the same radius

(2) Zr and Nb have similar oxidation state

(3) Zr and Hf have about the same radius

(4) Zr and Zn have the same oxidation state

FB0100

4. Which one of the following nitrates will leave behind a metal on strong heating ?

(1) Copper nitrate (2) Manganese nitate

(3) Silver nitrate (4) Ferric nitrate

DB0101

5. In which of the following oxidation state of Mn is six

(1) K_2MnO_4 (2) MnO_2

(3) $KMnO_4$ (4) MnO_4^{-1}

DB0103

6. Which has maximum magnetic moment :-

(1) Co^{+2} (2) Mn^{+3}

(3) Mn^{+2} (4) Cr^{+3}

DB0104

7. Which of the following element has maximum composition in misch metal ?

(1) La (2) Fe (3) Ce (4) Lu

FB0169

8. Which of the following lanthanoid ion has maximum spin magnetic moment ?

(1) Ce^{+3} (2) Pr^{+3} (3) Nd^{+3} (4) Pm^{+3}

FB0170

9. Which is related with lanthanoid contraction ?

(1) Radius of 4d & 5d series elements of a group is almost same

(2) IE of 5d series elements is very high as comapred to 4d series elements of a group

(3) Density of 5d series elements is abnormally high in its own group.

(4) All

FB0171

10. When hydrogen peroxide is added to acidified potassium dichromate, a blue colour is produced due to formation of

(1) CrO_3 (2) Cr_2O_3

(3) CrO_5 (4) CrO_4^{2-}

DB0109

11. The actinoids exhibit more number of oxidation states in general than the lanthanoids. This is because

(1) due to less energy difference between 6d & 5f

(2) the 5f orbitals are more buried than the 4f orbitals

(3) there is a similarity between 4f and 5f orbitals in their angular part of the wave function

(4) the actinoids are more reactive than the lanthanoids.

FB0111

12. In which of the following oxy anion all X–O bonds are not equivalent. (X – Central atom)

(1) MnO_4^- (2) CrO_4^{-2} (3) MnO_4^{-2} (4) CrO_7^{-2}

DB0172

13. When Cu^{+2} solution reacts with excess of KI, then which option is incorrect ?

(1) Insoluble Cu_2I_2 is obtained

(2) I_2 is liberated

(3) CuI_2 is formed

(4) It is a redox reaction between Cu^{+2} & I^-

DB0173

14. When MnO_2 is heated with fused KOH then the product obtained is

(1) $KMnO_4$, purple

(2) K_2MnO_4, green

(3) MnO_2, black

(4) Mn_2O_3, brown

DB0174

15. Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is **incorrect**?
(1) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.
(2) Ferrous oxide is more basic in nature than the ferric oxide.
(3) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.
(4) Ferrous compounds are less volatile than the corresponding ferric compounds.

DB0115

16. Which of the following metal cation is reduced from its higher oxidation state (+2) to (+1) by both KI solution and excess of KCN solution ?

(1) Zn^{2+} (2) Hg^{2+} (3) Cu^{2+} (4) None

DB0117

17. Cr^{2+} is reducing in nature because–
(1) In Cr^{2+}, configuration changes from d^4 to d^3 to acheive half filled t_{2g}.
(2) Cr^{2+} gain an electron to acheive d^5 configuration
(3) Cr^{2+} give an electron to acheive d^5 configuration
(4) In Cr^{2+} configuration changes from d^4 to d^3 to acheive half filled d-subshell.

DB0119

18 Which among the following order of oxidising power is correct–

(1) $CrO_4^- < MoO_4^- < WO_4^-$

(2) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$

(3) $CrO_3 < MnO_2 < Fe_2O_3$

(4) $Pb^{4+} < Sn^{+4} < C^{4+}$

DB0121

19 Which of the following reaction is not corrrect ?

(1) $Zn + \text{dil } H_2SO_4 \longrightarrow ZnSO_4 + H_2$

(2) $Cu + \text{dil } H_2SO_4 \longrightarrow CuSO_4 + H_2$

(3) $CuSO_4 + KI \longrightarrow K_2SO_4 + Cu_2I_2 + I_2$

(4) $Na_2Cr_2O_4 + H_2SO_4 \longrightarrow Na_2SO_4 + Na_2Cr_2O_7$

DB0175

EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**4**	**3**	**3**	**1**	**3**	**3**	**4**	**4**	**3**	**1**	**4**	**3**	**2**	**1**
Que.	16	17	18	19											
Ans.	**3**	**1**	**2**	**2**											